

CANADIAN OLYMPIC PARK UPPER, AB, Canada

WMO: 712350

Lat: 51.076N

Lon: 114.212W

Elev: 1235

StdP: 87.34

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-24.1	-30.6	0.3	-27.0	-27.3	0.4	-23.7	11.3	0.8	10.0	0.9	2.5	280	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	28.3	15.3	26.3	14.7	24.4	14.1	17.4	24.3	16.2	23.4	15.2	22.1	3.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	12.4	20.0	13.6	11.3	18.7	12.5	10.5	17.8	53.9	24.2	50.2	23.5	47.0	22.3	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.4	7.4	DB	-29.8	31.8	3.5	1.9	-32.3	33.2	-34.3	34.3	-36.3	35.3	-38.8	36.6	(4)
(5)			WB	-30.0	19.5	3.3	2.5	-32.4	21.3	-34.3	22.8	-36.2	24.2	-38.6	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-6.0	-5.5	-1.6	4.1	9.5	13.2	16.7	15.9	11.4	5.3	-1.3	-5.5	(6)	
	DBStd	10.29	8.97	8.09	7.81	5.43	4.87	3.51	3.32	3.65	5.00	5.31	7.27	7.82	(7)	
	HDD10.0	2582	496	434	359	187	68	10	1	3	42	161	341	479	(8)	
	HDD18.3	5003	754	667	617	427	274	159	71	91	212	405	590	738	(9)	
	CDD10.0	666	0	0	1	10	53	105	209	187	84	15	1	0	(10)	
	CDD18.3	46	0	0	0	0	1	4	21	16	4	0	0	0	(11)	
	CDH23.3	623	0	0	0	2	27	50	264	212	63	4	0	0	(12)	
	CDH26.7	136	0	0	0	0	2	8	65	48	11	2	0	0	(13)	
(14) Wind	WSAvg	3.4	3.5	3.3	3.5	3.6	3.6	3.4	3.2	3.0	3.1	3.5	3.4	3.5	(14)	
(15) Precipitation	PrecAvg	457	13	10	18	30	70	101	67	58	42	18	16	14	(15)	
	PrecMax	625	26	24	50	79	125	303	154	106	104	43	42	38	(16)	
	PrecMin	308	2	1	4	8	27	32	21	10	8	2	0	3	(17)	
	PrecStd	65	6	6	10	18	30	52	36	26	23	10	11	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	11.9	16.0	21.7	26.3	27.9	31.3	30.6	28.3	21.7	15.3	11.1	(19)	
		MCWB	4.5	4.3	6.4	9.1	12.3	15.5	15.9	15.1	13.4	10.4	6.7	3.4	(20)	
	2%	DB	8.7	9.0	12.8	18.2	23.4	24.6	28.4	28.0	25.1	18.1	11.9	8.2	(21)	
		MCWB	2.8	2.7	4.8	7.6	11.2	13.9	16.0	15.0	12.8	9.0	5.2	2.4	(22)	
	5%	DB	6.9	6.7	10.4	15.5	20.7	22.3	26.3	25.7	22.5	15.1	9.7	6.3	(23)	
		MCWB	1.8	1.3	3.5	6.3	10.0	12.9	15.6	14.6	12.0	7.3	3.8	1.2	(24)	
	10%	DB	5.1	4.6	8.1	12.6	18.3	20.4	24.1	23.6	19.9	13.0	7.5	4.4	(25)	
		MCWB	0.6	-0.2	2.2	4.7	9.0	11.9	15.0	14.2	10.9	6.1	2.3	0.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	4.8	7.0	9.8	13.3	17.0	19.2	18.7	15.2	11.4	7.6	4.5	(27)	
		MCDB	10.5	10.9	13.9	19.4	22.8	25.1	25.2	24.4	24.2	20.2	13.9	9.7	(28)	
	2%	WB	3.1	2.9	5.3	8.1	12.0	15.2	17.8	16.9	13.8	9.5	5.6	2.8	(29)	
		MCDB	8.1	8.3	12.2	16.9	21.1	22.2	24.5	23.7	22.6	16.8	11.3	7.7	(30)	
	5%	WB	2.0	1.5	3.8	6.6	10.8	14.0	16.7	15.8	12.7	7.9	4.1	1.3	(31)	
		MCDB	6.5	6.4	9.8	14.6	18.9	20.3	23.9	23.1	20.8	14.3	9.3	6.0	(32)	
	10%	WB	0.7	0.0	2.4	5.2	9.7	12.8	15.7	14.8	11.4	6.6	2.5	0.0	(33)	
		MCDB	5.0	4.2	7.8	11.8	17.1	18.8	22.6	21.9	18.7	12.2	7.1	4.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.7	10.0	10.6	11.4	10.8	12.2	12.2	11.4	10.0	8.7	8.5	(35)	
		MCDBR	10.2	11.7	13.0	14.8	15.2	13.9	15.4	15.6	15.6	13.2	11.8	10.1	(36)	
	5% WB	MCWBR	6.8	7.3	7.5	7.7	7.1	6.7	6.8	6.8	7.2	7.1	7.2	6.7	(37)	
		MCWBR	10.0	11.7	12.8	14.1	13.6	12.5	13.5	13.5	14.3	12.9	11.8	10.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.246	0.265	0.275	0.329	0.336	0.340	0.335	0.350	0.306	0.274	0.249	0.233	(40)	
		taud	2.439	2.449	2.489	2.338	2.387	2.412	2.427	2.397	2.491	2.564	2.511	2.425	(41)	
	Ebn at Noon		814	888	942	912	919	915	913	880	891	869	809	775	(42)	
		Edh at Noon	56	73	86	112	113	111	107	105	85	65	52	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.13	2.09	3.34	4.62	5.42	5.77	6.40	5.26	3.71	2.29	1.22	0.88	(44)	
	RadStd		0.08	0.15	0.24	0.33	0.43	0.44	0.42	0.37	0.41	0.23	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (65 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page